

Inter-modularity: Some Special Properties of Certain Quantifiers in German, With Focus on Uninflected *Ganz* and *Halb**

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This paper focuses on the special properties of *ganz* ‘whole’ and *halb* ‘half’ in German. It is observed that they do not have an inflection in the singular when a determiner is absent. Discussing other properties, it is documented that the nominals involving these quantifiers are situated between regular syntactic constructions and regular morphological formations. Given that the head noun must involve a proper name, it is proposed that the proper name incorporates into the quantifier as an instance of N-raising. At the end of the paper, some other exceptional quantifiers are briefly discussed.

1. Introduction

Adjectival quantifiers have an inflection when a determiner precedes. This is illustrated in the (a)-type of examples below, where the left column shows cases in the singular, and the right column documents instances in the plural. Interestingly, Roehrs (2026) observes that these quantifiers may also have an inflection when a determiner is absent, at least optionally as in the (b)-type of examples below. In the latter scenario, the inflection is obligatory in cases like (1b-b’) but optional in instances like (2b-b’) (for details, see the aforementioned paper):

- | | | | | |
|-----|----|--|-----|---|
| (1) | a. | das sämtlich*(-e) Obst
the all-INFL fruit | a’. | die sämtlich*(-en) Äpfel
the all-INFL apples |
| | b. | sämtlich*(-es) Obst
all-INFL fruit | b’. | sämtlich*(-e) Äpfel
all-INFL apples |
| (2) | a. | das viel*(-e) / wenig*(-e) Obst
the much-INFL / little-INFL fruit | a’. | die viel*(-en) / wenig*(-en) Äpfel
the many-INFL / few-INFL apples |
| | b. | viel(?-es) / wenig(?-es) Obst
much-INFL / little-INFL fruit | b’. | viel(-e) / wenig(-e) Äpfel
many-INFL / few-INFL apples |

There are only two exceptions to this generalization: The quantifiers *ganz* ‘whole’ and *halb* ‘half’ cannot have an inflection in the singular when a determiner is absent. This paper focuses on the special properties of these two elements but briefly comments at the end on some other quantifiers that are also exceptional (but in different ways): *lauter* ‘nothing but’, *welch-* ‘some’, and *meist-* ‘most’.

I show that nominals involving *ganz* and *halb* are situated between regular syntactic constructions and regular morphological formations – they seem to showcase inter-modularity. Crucially, these quantifiers can only combine with proper names. Referencing phrasal proper names like *Neues Ägyptisches Museum* ‘New Egyptian Museum’, I propose that *ganz* and *halb* are operators similar to *neu-* ‘new’. Unlike *neu-*, they are located in a head position. I propose that the proper name incorporates into the quantifier as an instance of N-raising (cf. Longobardi 1994).

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The paper is organized as follows. First, I present the data involving *ganz* and *halb*. Then, I offer my proposal in Section 3. Section 4 briefly discussed some other exceptional quantifiers. A short conclusion closes the paper.

2. Special Properties of *ganz* and *halb*

I start by discussing the special properties of *ganz* ‘whole’ and *halb* ‘half’, the main focus of this paper. After the presentation of the data, I summarize the properties and offer my proposal in the next section.

2.1. Special Properties: No Inflection

The quantifying words *ganz* ‘whole’ and *halb* ‘half’ exhibit special properties in the singular when no determiner is present: They cannot have an inflection (3a). By contrast, when a determiner like a definite article is present and in the plural more generally, all the forms of these elements have inflections and pattern in the regular way; consider (3b) and (4a-b):

- (3) a. *ganz*(*-es) / *halb*(*-es) Europa
 whole / half Europe
 b. *das ganz-e* / *halb-e* Europa
 the whole-INFL / half-INFL Europe
- (4) a. *ganz-e* / *halb-e* Äpfel
 whole-INFL / half-INFL apples
 b. *die ganz-en* / *halb-en* Äpfel
 the whole-INFL / half-INFL apples

The uninflected cases of *ganz* and *halb* in (3a) exhibit some other special properties.

2.2. Special Properties: Restricted Interpretation

Interestingly, *ganz* has different interpretations depending on the (in-)definiteness of and stress inside the hosting noun phrase (as far as I am aware, *halb* does not involve such differences in meaning). These readings are summarized below and illustrated further down:¹

- (5) *Interpretations:*
- a. Definite contexts:
 - unstressed *ganz*: ‘entire/all’
 - stressed *ganz*: ‘entire/all’; ‘intact’
- b. Indefinite contexts:
 - unstressed *ganz*: ‘entire/all’ (but not ‘all’ in the plural); after numerals, also ‘intact’
 - stressed *ganz*: predominantly ‘intact’, but ‘entire/all’ does not seem impossible
 - *ganz* can also mean: ‘just/merely’

¹ Kallulli & Rothmayr (2008) discuss determiner doubling constructions of the form Det + *ganz* + Det Adj N. Here, *ganz* is an adverbial element meaning ‘quite’. These cases will not be relevant here.

In the gloss lines below, I render *ganz* as ‘whole’ throughout, providing the actual interpretation(s) in the translation lines three, four, and (sometimes) five of the relevant strings.

Starting with some background information, note that nuclear stress in sequences of “determiner + adjective + noun” falls on the head noun (Cinque 1993: 248, 255); that is, under normal circumstances, *ganz* and *halb* are not stressed when part of a noun phrase. I begin the current investigation by discussing *ganz* in combination with common nouns and then turn to *ganz* in combination with proper names.

In definite contexts, and when *ganz* is not stressed, this element means ‘entire/all’:

- (6) a. Das ganz-e Auto kostet 30.000 Euro.
the whole-WK car costs 30,000 Euros
‘the entire car’
?‘the intact car’
b. Die ganz-en Autos standen auf dem Parkplatz.
the whole-WK cars were on the parking.lot
‘all (of) the cars’
?‘the intact cars’

Another possible reading in the singular is ‘all of’ (below is a googled example):

- (7) **Das ganze Auto** wurde untersucht und ...
the whole car was examined and
‘all of the car’

Proceeding more tentatively for a moment, it seems that ‘all of’ zooms in on the individual components of the entity under discussion, whereas ‘entire’ “zooms out” to the entity as a whole.

If *ganz* is stressed, it can mean both ‘entire/all’ and ‘intact’ (capitalization indicates stress):

- (8) das GANZE Auto, nicht das HALBE/KAPUTTE
the whole car not the half.one /broken.one
‘the ENTIRE car’
‘the INTACT car’

Note in passing that *ganz* in the sense of ‘entire/all’ can be combined with the quantifying element *alle* ‘all’ (9a), including in contexts of quantifier float (9b-d) (all are googled examples):

- (9) a. all die ganzen Jahre/Verbrechen/Menschen
all the whole years/crimes/people
b. **Die ganzen Angestellten** waren **alle** sehr nett und hilfsbereit.
the whole employees were all very polite and helpful
b’. **Die ganzen kleinen Städte hier** waren **alle** mega schön geschmückt.
the whole small towns here were all super nicely decorated
b’’. Aber **die ganzen Weichteile im Körper** waren ja **alle** am Verwesen.
but the whole soft.parts in.the body were PRT all at rotting

In indefinite contexts, and when *ganz* is not stressed, the interpretation of this element is restricted: It means ‘just/merely’ in the singular and ‘entire’ in the plural. In fact, Pafel (1994: 239 fn. 4) points out that plural *ganze Briefe* is unacceptable with the interpretation ‘all letters’, but fine with other interpretations. As implied by Pafel, this holds more generally ((10b) is a googled example)

- (10) a. Ein ganz-es Auto wurde verkauft.
 a whole-ST car was sold
 ‘just/merely one car’
 ?‘an intact car’
- b. In meinem toten Winkel verschwinden **ganze Autos**.
 in my blind spot disappear whole-ST cars
 ‘entire cars (but not necessarily all of them)’
 ?‘intact cars’

Another possible reading in the singular is ‘entire’ ((11a) is a googled example). A further possible reading in the plural is ‘just/merely’ (11b-b’). Also, after numerals, *ganz* can also mean ‘intact’:

- (11) a. ein Autoteil innerhalb von nur einer Woche gegen **ein ganz-es Auto**
 a car.part within of only one week for a whole-ST car
 einzutauschen
 swap
 ‘an entire car’
- b. ganz-e vier Autos
 whole-ST four cars
 ‘just/merely four cars’
- b’. vier ganz-e Autos
 four whole-ST cars
 ‘just/merely four cars’
 ?‘four entire cars’
 ‘four intact cars’

If *ganz* is stressed, it preferably means ‘intact’, but the meaning ‘entire/all’ does not seem impossible:

- (12) ein GANZES Auto, nicht ein HALBES/KAPUTTES
 a whole car not a half.one /broken.one
 %‘an ENTIRE car’
 ‘an INTACT car’

With this in place, I turn to *ganz* in combination with proper names like *Europe*. Here, *ganz* behaves differently.

As is well known, proper names are definite in interpretation, even when no article is present. Crucially, the interpretation of *ganz* is quite restricted. Independent of stress, *ganz* only means ‘entire/all’ (but not ‘intact’):

- (13) a. ganz Europa
 whole Europe
 ‘all of Europe’
 #‘intact Europa’
 b. GANZ Europa ist davon betroffen
 whole Europe is by.that impacted
 ‘ALL of Europe’
 #‘INTACT Europe’
 c. nur GANZ Europa kann weiterbestehen
 only whole Europe can continue.exist
 ‘ALL of Europe’
 #‘INTACT Europe’

Interestingly, the interpretation ‘intact’ is possible when a determiner is present, and *ganz* is stressed:

- (14) a. das GANZ-E Europa, nicht das HALBE/KAPUTTE
 the whole-WK Europe not the half.one/broken.one
 ‘ALL of Europe’
 ‘the INTACT Europe’
 b. nur ein GANZ-ES Europa kann weiterbestehen, nicht ein KAPUTTES
 only a whole-ST Europe can continue.exist not a broken.one
 ‘only an INTACT Europe’

To take stock, while this may not be an exhaustive discussion of the different readings of *ganz* in the various contexts, it seems clear that *ganz* without a determiner differs in interpretation from *ganz* with a determiner. Specifically, with the possible exception of contexts involving numerals, *ganz* in the sense of ‘intact’ requires the presence of a determiner (and stress). This means that *ganz* itself is not a determiner (which is particularly clear with *ganz* in the meaning of ‘intact’). Furthermore, when inflected *ganz* is stressed, it can typically mean ‘entire/all’ in definite contexts only (e.g., after the definite article). Since this interpretation is also possible with stressed uninflected *ganz*, this is evidence that *ganz Europa* is indeed definite in interpretation.

2.3. *Special Properties: Not Determiners Themselves*

There is independent evidence that *ganz* (and *halb*) are not determiners: The directional preposition ‘to’ in English has two counterparts in German: one that takes a complement with a definite article (*in*) and one that takes a complement without a definite article (*nach*). As is well known, *Schweiz* takes an article in German (15a-b). As seen in Section 2.1, this is different for *Europa*: The latter takes an article when inflected *ganz* is present but not when *ganz* is uninflected or absent. In the first case, the noun phrase is preceded by *in* (15c), in the second case by *nach* (15d). By contrast, note that quantificational determiners like *jeder* ‘every’ occur with *in* (15e):

- (15) a. Die Waren gehen in die (ganz-e) Schweiz.
 the goods go to the (whole-WK) Switzerland

- b. * Die Waren gehen nach (der) Schweiz.
the goods go to (the) Switzerland
- c. ? Die Waren gehen ins ganz-e Europa.
the goods go to.the whole-WK Europe
- d. Die Waren gehen nach (ganz) Europa.
the goods go to (whole) Europe
- e. Die Waren gehen in jedes Land.
the goods go in every country

Comparing (15a), (15d), and (15e), it is clear that *ganz* (and *halb*) is not a determiner element.

2.4. *Special Properties: Only Proper Names*

Continuing with the singular articleless cases, *ganz* and *halb* are not possible with common nouns such as count nouns, mass nouns, or common nouns denoting unique entities:

- (16) a. * ganz Auto/Obst/Sonne
whole car/fruit/sun
- b. * halb Auto/Obst/Sonne
half car/fruit/sun

Turning to proper names, note that certain instances of such elements are quantifiable: The entities they denote can be divided into smaller parts, such that the smaller part can be referred to by a quantifier and the proper name (e.g., *halb Europa* ‘half (of) Europe’).² Observe now that quantifiable proper names such as geographical names for continents, countries, and cities are possible, provided they do not take an article (17a) vs. (17b). Note that (17c) involves a country name that is clearly in the plural. Also, street names and names for rivers and political parties are not possible (17d). All the cases in (17b-d) involve an article:

- (17) a. ganz Europa/Deutschland/Berlin/Den Haag
whole Europe/Germany/Berlin/The Hague
- b. * ganz Schweiz/Türkei/USA/Niederlande
whole Switzerland/Turkey/USA/Netherlands
- c. * ganz Vereinigt-e(-n) Staaten von Amerika
whole United-ST(-WK) States of America
- d. * ganz Mozartstraße/Rhein/SPD
whole Mozart.street/Rhine/SPD

The city name *Den Haag* in (17a) above does not involve an article in German. This is confirmed in Saxon Genitive contexts (18a), when compared to *die Schweiz* ‘(the) Switzerland’, which clearly does (18b-b’). Similar facts hold for proper names that do not end in a sibilant (18c). Given (18b-c), it seems clear that proper names with an obligatory article cannot form the possessor in Saxon Genitives (also Roehrs 2020), corroborating that *Den* is not interpreted as an article in German:

² This is somewhat similar to mass nouns (which are not entities).

- (18) a. Den Haags Bevölkerung
The Hague's polulation
b. ?* de-r Schweiz Bevölkerung
the-GEN Switzerland population
b'. * Schweiz'/Schweizes Bevölkerung
Switzerland's population
c. ?? Türkeis Bevölkerung
Turkey's population

As Nübling et al. (2015: 83) point out, feminine country names require the presence of an article (cf. (17b) above), neuter country names do not tolerate its presence (cf. (17a) above), and masculine country names often omit the article: *(der) Iran*, *(der) Irak*. If *ganz* and *halb* only combine with articleless proper names, then we predict that masculine proper names can also occur with *ganz* and *halb*. This is indeed borne out: *ganz Iran*, *halb Irak*. This means that *ganz* and *halb* can combine not only with neuter singular proper names but also singular proper names of other genders.

Furthermore, certain names for music bands are possible:

- (19) a. Halb Kool & the Gang war krank.
half Kool & the Gang was sick
b. Ganz Maroon 5 war gekommen.
whole Maroon 5 had come

This also includes *Hootie & the Blowfish* and other bands that involve several members. Below are some googled examples involving the name of a German band:

- (20) a. kannst du auch mal Edits über **ganz Tokio hotel**   machen
can you also PRT edits about whole Tokio Hotel make
b. ich will nicht mit **halb Tokio Hotel** durch die Stadt rennen
I want not with half Tokio Hotel through the city dash

By contrast, singular personal names are not possible – they are typically not devisable, and thus not quantifiable:³

- (21) a. * ganz Maria
whole Mary
b. * halb Peter
half Peter

Finally, phrasal proper names as in (22) are not possible either, where (22b) might be out independently due to the presence of an adjective (see Section 2.6):

³ This may hold in regular contexts only: The case below does not seem impossible:

(i) ? halb Peter war dafür, halb Peter war dagegen
half Peter was in.favor half Peter was against

- (22) a. * ganz Hochschule für Bildende Künste Dresden
 whole College for Creative Arts Dresden
 b. * ganz Deutsche Bahn
 whole German Railway

To sum up, uninflected *ganz* and *halb* combine with (singular) proper names that do not take an article and are quantifiable.

2.5. *Special Properties: No Determiner(-like) Elements*

Above, we saw that uninflected *ganz* and *halb* in the singular occur without a preceding determiner. As already seen in (3b) above, the definite article is possible, provided *ganz* and *halb* take an inflection (23a). Saxon Genitives seem to be possible in front as well, but again, *ganz* and *halb* must take an inflection ((23b-c) sound possible, but I have not found any examples through google):

- (23) a. das ganz*(-e) / halb*(-e) Europa
 the whole-WK / half-WK Europe
 b. Nikolais/Katharinas/Putins ganz*(-es) Russland/Rußland
 Nikolai's/Katharina's/Putin's whole-ST Russia
 c. Merkels ganz*(-es) / halb*(-es) Deutschland
 Merkel's whole-ST / half-ST Germany

The co-occurrence of these two properties (i.e., presence of a determiner(-like) element and inflection on *ganz* and *halb*) makes these cases instances of regular noun phrases and thus irrelevant to the current discussion. Importantly, uninflected *ganz* and *halb* cannot be preceded by any type of determiner.

2.6. *Special Properties: Cohesion between ganz/halb and the Proper Name*

Regular adjectives cannot intervene. This means that uninflected *ganz* and *halb* in combination with a proper name involve a cohesive, inseparable combination. Note that it is not generally impossible that an uninflected quantifier can be followed by an adjective, comparing (24a) to (24c):

- (24) a. * ganz / halb groß-e(-s) Europa
 whole / half big-WK(-ST) Europe
 b. ganz-e / halb-e frisch-e Äpfel
 whole-ST / half-ST fresh-ST apples
 c. viel interessant-es Zeug
 much interesting-ST stuff

The cohesion is also reflected by the impossibility of *halb Europa* undergoing Split Topicalization, comparing (25a) to (25b). Note that *halb* cannot (easily) be an adverbial element

after *nicht* ‘not’ (25c), that stranded material, in general, does not have to bear an inflection (25d), and that Split Topicalization is not impossible with definite noun phrases (25e-f):⁴

- (25) a. Ich habe nicht HALB Europa besucht, sondern GANZ Europa.
 I have not half Europe visited but whole Europe
 b. * **Europa** habe ich nicht **halb** besucht.
 Europe have I not half visited
 c. ?* Ich habe das nicht halb gemacht.
 I have that not half done
 d. **Brötchen** habe ich nicht **zwei** gegessen, sondern drei.
 rolls have I not two eat but three
 e. ?? **Europa** habe ich eigentlich nur **das halbe** besucht.
 Europe have I actually only the half visited
 f. **Brötchen** habe ich immer nur **diese** gegessen.
 rolls have I always only these eaten

Unlike *halb* ‘half’, *ganz* can be an adverbial element meaning ‘entirely’ after the negation (26a), thus giving a different, irrelevant result under topicalization (26b):

- (26) a. Ich habe das nicht ganz erledigt.
 I have that not entirely completed
 b. Europa habe ich nicht ganz besucht.
 Europe have I not entirely visited

Crucially, uninflected *ganz* and *halb* have to be adjacent to the following proper name.

2.7. Special Properties: Compound-type Formation

Intriguingly, *ganz* and *halb* can be part of complex Saxon Genitives. To see this, I briefly consider the so-called Genitive Rule in German (e.g., Gallmann 2018). According to this rule, genitives must involve an inflected prenominal modifier as part of the genitive phrase, unless the genitive phrase itself is a Saxon Genitive.

In more detail, the workings of the Genitive Rule are illustrated in the singular in (27) and in the plural in (28) ((28d) is provided with Haider’s 1992: 330 judgments):

- (27) a. * der Verkauf Stoff-es
 the sale fabric-GEN
 b. der Verkauf lila^{??}(-nen) Stoff-es
 the sale purple-GEN fabric-GEN
- (28) a. * der Verkauf Autos
 the sale cars

⁴ This argument comes with a pinch of salt, as the crucial example in (25b) might be out independently: As shown in (25d), a correction (introduced by *sondern* ‘but’) is present, if *nicht* ‘not’ occurs before the quantifier in the first part of the sentence. Now, if a correction is added to (25b) (e.g., *sondern ganz* ‘but entire(ly)’), then *ganz* and *halb* get an (irrelevant) adverbial reading.

- b. der Verkauf lila^{??}(-ner) Autos
the sale purple-GEN cars
- c. der Verkauf zwei*(-er) Autos
the sale two-GEN cars
- d. * der Verkauf Peters Häuser
the sale Peter's houses

Crucially, Saxon Genitives are fine without inflected prenominal modifiers:

- (29) a. die Bücher Peter Müllers
the books Peter Müller's
- b. die Geschichte Deutschlands
the history Germany's

Consider *ganz Europa* in genitive contexts. Besides the suffix *-s* on the head noun (30), coordination in (31) clearly indicates that Saxon Genitives are indeed dependents in the genitive (all are googled examples):

- (30) a. Eine de-r meistbesuchten [Sehenswürdigkeiten **ganz Europas**]
one the-GEN most.visited sights whole Europe's
- b. Die Esche ist im [Tief- und Hügelland fast **ganz Europas**] von Natur
the ash-tree is in.the low- and hilly.country almost whole Europe's by nature
aus verbreitet
PRT spread
- c. innerhalb **ganz Europas**
inside whole Europe's
- (31) a. daß die historische [Entwicklung de-r Union und **ganz Europas**]
that the historical development the-GEN Union and whole Europe's
- b. Grund genug um mit der UNITED-Brille über die [Rasen **halb Europas**, de-s Nahen Osten-s] bis nach Übersee zu schweifen.
reason enough in.order.to with the UNITED-glasses across the lawns half Europe's the-GEN Middle East-GEN all.the.way to overseas to roam
- c. Als Bundesminister der Finanzen hat er in herausfordernden Jahren nicht nur
as federal.secretary of finance has he in challenging years not only
für die [Fiskalpolitik Deutschlands, sondern **ganz Europas**] Maßstäbe gesetzt.
for the fiscal.policy Germany's but whole Europe's standards set

Now, if *ganz* and *halb* were regular adjectives, we would expect these two elements to yield bad strings when they are uninflected. However, the examples are fine. Thus, these two elements seem to be part of complex Saxon Genitives but not regular DPs in the genitive. In other words, these instances seem to behave more like some kind of compound formation.

Possessives can also be in prenominal position, provided there is no inflected element present, as shown in (32a) vs. (32b-c) (for (32b), see Gallmann 1997: 81 and Zifonun et al. 1997: 2064):⁵

⁵ There are two types of exception to the requirement of Saxon Genitives not having an inflection. The first involves *Herr* 'mister' and less frequently *Kollege* 'colleague' and *Genosse* 'comrade' (Zifonun et al. 1997: 2046-47). The

- (32) a. ?? de-s Peters Auto
the-GEN Peter's car
b. Klein-Marias Fahrrad
Little-Maria's bicycle
c. Häns-chen kleins Rucksack
Häns-little small's knapsack

Ignoring very similar repeats in each case, I looked through all 340 google hits of *ganz Europas* and all 300 google hits of *halb Europas* and identified four prenominal examples ((33a) is the title of written work, (33b) involves lyrics of a song; and (33c-d) are from regular prose):⁶

- (33) a. Spanien ist [**ganz Europas** Drogen-Dealer]
Spain is whole Europe's drug-dealer
b. Aber der Alte wollt' nicht so, er bracht' ihn um, Und [**halb Europas**
but the old.one wanted not thus he killed him PRT and half Europe's
Samen], Mann für Mann.
seeds man by man
c. und diggt in seiner Freizeit [**halb Europas** Plattenläden] leer
and digs in his spare.time half Europe's record.stores empty
d. Als die wundersame Reise de-s FK Bodö/Glimt durch [**halb Europas**
when the wondrous journey the-GEN FK Bodö/Glimt through half Europe's
Fußball] ein jähes Ende fand, ergriff ...
soccer a sudden ending found, took

Thus, although rare, prenominal uses of the relevant Saxon Genitives are possible as well. Again, rather than regular DPs in the genitive, *ganz* and *halb* are part of complex Saxon Genitives, which seem to involve some kind of compound element.

2.8. *Special Properties: Stress*

Perhaps surprisingly, stress is on the following proper name, a typical feature of a regular syntactic construction, but not of a regular compound noun (again, capitalization indicates stress):

second type of exception consists of an appositive involving a nominalized (adjectival) participle that can appear before the proper name (Duden 1995: 244):

- (i) a. Herr(n) Müllers Auto
Mr(-INFL) Müller's car
b. Vorsitzend-en Schmidts Ausführungen
chairman-INFL Schmidt's remarks

Note that the nouns in the first case are all weak masculine nouns but that not all such nouns can have an inflection in Saxon Genitive contexts (e.g., *Professor-(*en) Müllers Auto* 'professor Müller's car'). This may have to do with the fact that the latter case adds a schwa, that is, another syllable in the genitive form: *Professo.ren*. This seems to be confirmed by the markedness of the disyllabic genitive form of *Herr*: ??*Herren Müllers Auto*.

⁶ I identified a fifth example, also the title of written work. Despite the round brackets in the original, this example is actually structurally and lexically ambiguous, indicated by square brackets below:

- (i) (Noch nicht [**ganz** **Europas** führende Versicherungsplattform]
still not yet)[whole Europe's leading insurance.portal]

- (34) a. ganz EuROpa
whole Europe
b. halb RUSSland
half Russia

The cases above pattern with (35a) below, which involves another combination of uninflected adjective + proper name, but not (35b) below, which involves a compound:

- (35) a. Klein-MaRIa
Little-Maria
b. KLEINstadt
small.town

Similarly, as with *ganz* and *halb*, an article is not possible with *Klein-Maria* (36a), again unlike with regular compounds like *Kleinstadt* (36b):

- (36) a. * die Klein-Maria
the Little-Maria
b. die Kleinstadt
the small.town

The stress patterns and the distribution of the article above are more general: Besides *klein* ‘small’, this also holds with *groß* ‘big’, *neu* ‘new’, and *alt* ‘old’, where the proper names in (37) are stressed on the non-initial element, but the compounds in (38) on the initial one:⁷

- (37) a. KleinAsien, KleinBLITtersdorf, KleinMACHnow, KleinOSTheim
Minor.Asia Small.Blittersdorf, etc.
b. GroßbriTANNien, GroßRÖHRSDorf, GroßWALDstadt, Groß SCHÖnebeck
Great.Britain Great.Röhrsdorf, etc.
c. NeuBRANdenburg, NeuKÖLN, NeuSCHWANenstein, NeuSTRElitz
New.Brandenburg, etc.
d. AltkasTilien, AltGLIEnicke. AltHAgEn, Alt RupPIN
Old.Castille Old.Glienicke, etc.
- (38) a. KLEINaktionär, KLEINbau, KLEINbürger, KLEINGartenanlage
small.investor, small.construction small.citizen small.garden.grounds
b. GROSSartiges, GROSSbrand, GROSSEinkauf, GROSSenkel
great.things great.fire great.shopping great.grandson
c. NEUBau, NEUbeginn, NEUentwicklung, NEULand
new.construction new.start new.development new.land
d. ALTbau, ALTbundeskanzler, ALTdeutsch, ALTsilber
old.construction old.chancellor old.German old.silver

Unlike common nouns, proper names can not only occur with uninflected adjectives, but they can also keep their original stress pattern.

⁷ There seem to be exceptions in both directions: *NEUstadt*, *ALTdorf* vs. *AltWEIbersommer*.

3. Analysis

I start with the summary of the special properties of *ganz* and *halb*. After briefly discussing and dismissing an alternative proposal, I offer my own analysis. Finally, I turn to some differences between *ganz Europa*, *Klein-Maria*, and *Großbritannien*.

3.1. Summary of the Special Properties

I summarize the properties of the special cases of *ganz* and *halb*, namely when these words occur in the singular and without a determiner:

(39) *Summary:*

- uninflected
- unstressed in neutral contexts (stress is on the following noun)
- cohesion between *ganz* and *halb* and the following noun
 - o adjectives cannot intervene
 - o Split Topicalization is impossible
- only proper names can follow, provided they
 - o are singular
 - o are quantifiable
 - o occur without a determiner
- Saxon Genitives (and other determiners) cannot precede
- restricted interpretation – even if stressed, *ganz* can only mean ‘entire/all’ (unlike inflected *ganz* with a determiner, which may be ambiguous)
- despite the last three (underlined) properties, *ganz* and *halb* are not determiners themselves
 - o complement of the preposition *nach* ‘to’
 - o part of prenominal and postnominal complex Saxon Genitives (rather than regular DPs in the genitive)

Given these properties, uninflected *ganz* and *halb* in combination with a proper name seem to showcase inter-modularity: These cases seem to be situated between regular syntactic constructions (word order, stress, only proper names without an article) and regular morphological formations (various restrictions such as lack of inflection and cohesion).

3.2. Proposal

I entertain two types of accounts. Briefly discussing and dismissing the first, I elaborate on the second.

Proposal #1: These combinations could involve copulative compounds. However, the latter only combine two likes and do allow an article (e.g., *der Drucker-Kopierer* ‘the printer-copier’). As such, this type of proposal is unlikely to be correct.

Proposal #2: I employ selection in a way to be made more precise below. To set the stage, Julien & Roehrs (2023: 41-42) propose for phrasal proper names in Norwegian as in (40a-b) that (weakly-inflected) *nye* ‘new’ is an operator (40c). It is a function that takes an existing name and

returns a new name (on the basis of the old existing name). It is a head projecting a functional phrase (FP), and it selects a DP involving proper names:

- (40) a. Ny-e Elses Blomster [store] (Norwegian)
 new-WK Else's Flowers
 'New Else's Flowers'
- b. Ny-e Grei-Ø Kafeteria [cafeteria]
 New-WK Nice-ST Cafeteria
 'New Nice Cafeteria'
- c. *nye*OP: ____DP_[+proper]

As to German, Julien & Roehrs (2023: 45) suggest that (strongly-inflected) *neu-* 'new' is also an operator, but it is in Spec,FP, and F selects a regular non-DP structure (e.g., AgrP) (note that some speakers allow the non-parallel weak ending *-e* on the second adjective, indicating that *neu-* is not a regular adjective here):

- (41) Neu-es Ägyptisch-es Museum [museum]
 New-ST Egyptian-ST Museum
 'New Egyptian Museum'

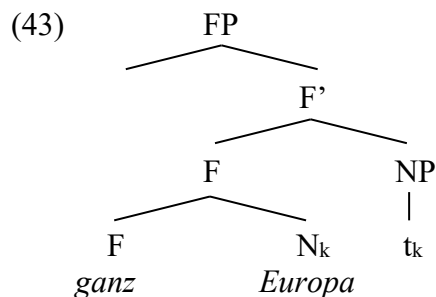
Notice that the different inflections on Norwegian *nye* and German *neu-* have an independent explanation (Roehrs & Julien 2014). More importantly, Julien & Roehrs (2023) point out that besides *neu-* 'new', there are a few other adjectives working this way: *alt-* 'old', *klein-* 'small', *groß-* 'great'.

The elements discussed in this paper are also operators, but they have possibly more progressed in their development. I take them to be heads as in Norwegian, but they select a NP (42a). The minimal structure they project is provided in (42b):

- (42) a. *ganz*OP: ____NP_[+proper]
- b. [FP F [NP N_[+proper]]] where F = *ganz, halb, neu, alt, klein, groß*

As also seen with the phrasal proper names above, there is a limited set of lexical items that belong to this group of operators – observe the overlap of the four last items in (42b) with the operators mentioned further above, the difference being that all operators discussed here are uninflected.

The derivation proceeds as follows where the proper name incorporates into F as an instance of N-raising:



This explains the word order, the cohesiveness, and the lack of inflection on *ganz* and *halb*. I assume that FP forms the possessor in Saxon Genitives but that a DP can presumably be projected in argument position of verbs, where F+N moves to D as an instance of N-to-D movement (cf. Longobardi 1994). These structural and derivational assumptions account for the absence of a determiner. Since (non-personal) proper names in the plural involve an article, they cannot combine with *ganz* and *halb*, accounting for the restriction to singular. Furthermore, common nouns do not undergo N-raising in German, explaining why *ganz* and *halb* can only occur with proper names.

While perhaps not very common, right adjunction to a higher head as in (43) above is familiar from Emonds' (1985: 207) and Abney's (1987: 287) proposal of indefinite pronoun constructions (44a) and from Pollock's (1989: 414-416) analysis of the French negation *ne ... pas* 'not' (44b):⁸

- (44) a. some+thing_i big t_i
 b. [T n'k+a_i] pas t_k t_i (French)
 NEG+has not

Also, the cases under discussion here might be relatable to cases of so-called Adjective Incorporation in Northern Swedish (Julien 2005: 58), where the adjective incorporates into the noun.⁹ This incorporation allows N-to-D movement, blocking the appearance of a free-standing determiner, comparing (45a) to (45b):

- (45) a. en gammal svart katt (Northern Swedish)
an old black cat
b. gamm-svart-katt-a
old-black-cat-DEF

Note that like above, incorporation occurs in definite contexts. Next, I briefly discuss some differences between *ganz Europa* and a few other, similar constructions.

3.3. Some Differences Between ganz Europa, Klein-Maria, and Großbritannien

First, *ganz* can be modified as seen below (the list of modifiers is taken from Pafel 1994: 256):

⁸ Note that the details of the analysis in (44b) depend on the specifics of the (clitic) movement of *ne*.

⁹ I am not certain as to whether or not incorporation of the adjective into the noun is the best analysis here. Having said that, note that Julien (2005: 62) suggests that this may involve the formation of a prosodic word.

- (46) a. nicht ganz Europa
not whole Europe
b. mindestens/höchstens/ungefähr /nur /wohl /vermutlich halb Europa
at.least /at.most /approximately/only/probably/presumably half Europe

Interestingly, *mehr als* ‘more than’ and *fast* ‘almost’ are special – they seem to be able to access elements inside more complex structures:

- (47) a. mehr als halb Europa
more than half Europe
b. mehr als das halb-e Europa
more than the half-WK Europe
c. ?? das mehr als halb-e Europa
the more than half-WK Europe
- (48) a. fast ganz Europa
almost whole Europe
b. fast das ganz-e Europa
almost the whole-WK Europe
b'. ? das fast ganz-e Europa
the almost whole-WK Europe

Furthermore, for a different set of data, Pafel (1994: 256) observes that modifiers can precede prepositions. This is also possible here (according to Pafel, the modifier moves from the quantifier up to the preposition):

- (49) a. fast in ganz Europa
almost in whole Europe
b. (?) fast im ganz-en Europa
almost in.the whole-WK Europe

This means that *ganz* can be modified.

By contrast, the adjectival element in other combinations of uninflected adjective + proper name (50a-b) or compounds more generally (50c) cannot be modified by a degree element:

- (50) a. * sehr Klein-MaRIa
very Little-Maria
b. * sehr KleinAsien
very Asia.Minor
c. * (der) sehr KLEINaktionär
the very small.investor

Second, determiners must occur when proper names have genitive dependents (51a). However, the combination of uninflected *ganz/halb* + proper name (51b-b') or uninflected *klein* + proper name (51c-c'') do not allow a determiner:

- (51) a. *(das) Großbritannien mein-er Kindheit
the Great.Britain my-GEN childhood
b. ?? Halb Europa mein-er Kindheit war Teil de-s Warschauer Pakt-es.
half Europe my-GEN childhood was part the-GEN Warsaw Pact-GEN
b'. Das halb-e Europa mein-er Kindheit war Teil de-s Warschauer Pakt-es.
the half-WK Europe my-GEN childhood was part the-GEN Warsaw Pact-GEN
c. * (die) Klein-Maria mein-er Kindheit
the Little-Maria my-GEN childhood
c'. die klein-e Maria mein-er Kindheit
the little-WK Maria my-GEN childhood
c''. (*die) Klein-Maria, die ich gestern gesehen habe
the Little-Maria who I yesterday seen have
(non-restrictive reading only)

Note that postnominal Saxon Genitives are not possible with *ganz Europa* either (notice that I did not find any googled examples for (52b-c), but (52c) sounds much better than (52b)):

- (52) a. das Deutschland Merckels
the Germany Merkel's
b. * ganz Deutschland Merckels
whole Germany Merkel's
c. das ganz-e Deutschland Merckels
the whole-WK Germany Merkel's

Given the discussion above, it seems clear that *ganz Europa* shares some properties with *Klein-Maria*, but the former also exhibits some difference to the latter. Both of these combinations of uninflected adjective + singular proper name are different from compounds like *Großbritannien*.

4. Special Properties of Some Other Quantifiers in German

Finally, I briefly point out some special properties of *lauter* 'nothing but', *welch-* 'some', and *meist-* 'most', which are different from *ganz* and *halb*. Most of the discussion is dedicated to the less oft-studied *lauter*.

4.1. *Lauter*

According to Eckardt (2006: 204), *lauter* is a determiner derived from an old adjective meaning 'pure'. In more detail, syntactically, Eckardt takes it to be a determiner(-like) word, but semantically, she argues that *lauter* has remained an adjective (her page 233). *Lauter* has a meaning along the lines of 'many' (but this component can disappear, her page 210) and 'only' (but this part can also vanish, her page 206). Here, *lauter* is translated as 'nothing but' throughout:

- (53) Die Maiers haben lauter Töchter.
the Maier-family have nothing.but daughters
'The Maier family have only daughters, and a fair number of them at that.'

Eckardt argues that *lauter* is a determiner syntactically (54a-b) (note, however, that these two distributions are independently out; for (54a), see Roehrs 2026). Also, observe that the use of *lauter* as an adjective as in (54c) is fine in older varieties of German:

- (54) a. * die / einige lauter Pfifferlinge
the / some nothing.but chanterelles
b. * lauter die / zwei Morchel
nothing.but the / two morels
c. * lauter-es Wasser
pure-ST water

Eckardt points out that *lauter* is not a focus particle, comparing (54b) above to (55a) below:

- (55) a. nur /bloß die Morchel
only/barely the morels
b. Hans hat nur /bloß/*lauter geschlafen
Hans has only/barely/nothing.but slept

Furthermore, she notes that *lauter* cannot occur in Split Topicalizations:

- (56) Steinpilze haben wir haufenweise/massig /*lauter gefunden.
penny.bun.bolets have we in.heaps /in.masses/nothing.but found

Without going into any details here, Eckardt (2006: 229-31) argues that Split Topicalizations involving *lauter* are ruled out by the semantics.

Referencing Eckardt (2006), Kobele & Zimmermann (2012: 242, 280) point out similar and some additional properties. Like Eckardt, they point out that *lauter* must occur without a determiner and that it cannot be inflected. In addition, they state that *lauter* cannot be modified and requires a nominal complement. Kobele & Zimmermann do not give examples of these properties, but the latter two could be illustrated as follows (note that (57b) involves a context different from Split Topicalizations):

- (57) a. * fast /sehr lauter Töchter
almost/very nothing.but daughters
b. * Familie Brandt hat nur wenige Töchter, aber Familie Maier hat lauter.
Family Brandt has only few daughters, but family Maier has nothing.but
c. ? Familie Brandt hat lauter Töchter, aber Familie Maier hat nur wenige.
Familie Brandt hat nothing.but daughters but family Maier has only few

Note, however, that *lauter* can be modified (58a) and that it can also function as a degree element (58b-b') (all are googled examples):

- (58) a. fast nur lauter Tuchmacher/Männer/alte Bekannte/Einser
almost only nothing.but cloth.workers/men/old acquaintances/A's (grades)

- b. vor lauter viel Arbeit/Eis/Spaß
because.of very? much work/ice/fun
- b'. lauter viele Dinge/Ideen/Gründe
very? many things/ideas/reasons

Presumably, *lauter* is a degree element in the latter cases.

4.2. *welch-*

The quantificational element *welch-* ‘some’ seems to be an indefinite pronoun, in that it cannot be followed by a noun (59a). In addition, this element seems to have some anaphoric properties, in that it requires an antecedent. This is illustrated by Split Topicalizations (59b). Intriguingly, if *irgend-* ‘any’ is prefixed to *welch-*, a noun becomes possible (59c) (Haider 1990: 108; Kobele & Zimmermann 2012: 265; for (59c), cf. Zifunon et al. 1997: 1941):

- (59)
- a. * Ich habe welches Brot gekauft.
I have some bread bought
 - b. Brot habe ich welches gekauft.
bread have I some bought
 - c. Ich habe dann einfach irgendwelches Brot gekauft.
I have then simply any.some bread bought

4.3. *meist-*

Finally, the superlative form of *viel* ‘much/many’ is (suppletive) *meist-*. Kobele & Zimmermann (2012: 254) point out that due to this morphological form, *meist-* must be preceded by a definite article. Note in this regard that the English counterpart *most* also involves a superlative form but does not have an article. However, the latter appears when a relative clause is added:

- (60)
- a. *(die) meisten Leute
the most people
 - b. the most I have ever seen

This concludes the brief survey of some other quantifiers that also exhibit some exceptional properties.

5. Conclusion

As documented above, certain quantifiers may show a number of exceptional properties. Focusing on uninflected *ganz* and *halb*, I demonstrated that these cases show properties of regular syntactic constructions and regular morphological formations. I proposed that these instances involve incorporation of the proper name into the quantifier as an instance of N-raising.

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